



ANDERSON SERAGOON JUNIOR COLLEGE
JC2 Preliminary Examinations 2019
Higher 3

H3 MATHEMATICS

9820/01

Paper 1

20 September 2019

3 hours

Additional Materials: Answer Booklet
 List of Formulae (MF26)

READ THESE INSTRUCTIONS FIRST

Write your name, class, syllabus and paper number on the answer booklet you hand in.
Write in dark blue or black pen on both sides of the answer booklet.
You may use an HB pencil for any diagrams or graphs.
Do not use staples, paper clips, glue or correction fluid.

Answer **all** the questions.

Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.

You are expected to use an approved graphing calculator.

Unsupported answers from a graphing calculator are allowed unless a question specifically states otherwise.

Where unsupported answers from a graphing calculator are not allowed in a question, you are required to present the mathematical steps using mathematical notations and not calculator commands.

You are reminded of the need for clear presentation in your answers.

At the end of the examination, ensure that you have submitted all your work.

The number of marks is given in brackets [] at the end of each question or part question.

This document consists of **5** printed pages and 1 blank page.

- 1 (a)** Show that $\gcd(a+b, a^2-ab+b^2) = \gcd(a+b, 3ab)$ for non-zero integers a and b .
If $\gcd(a, b) = 1$, show that $\gcd(a+b, a^2-ab+b^2)$ is either 1 or 3. [7]
- (b)** Show that $\gcd(n!+1, (n+1)!+1) = 1$, where n is a positive integer. [4]
- 2 (i)** It is given that all the terms in the two sequences $p_1, p_2, \dots, p_n$ and $q_1, q_2, \dots, q_n$ are non-zero, where $n \geq 1$. By considering the function $f(x) = \sum_{r=1}^n (p_r x - q_r)^2$, prove that
- $$\left(\sum_{r=1}^n p_r^2 \right) \left(\sum_{r=1}^n q_r^2 \right) \geq \left(\sum_{r=1}^n p_r \cdot q_r \right)^2,$$
- and that equality holds if and only if $\frac{q_1}{p_1} = \frac{q_2}{p_2} = \dots = \frac{q_n}{p_n}$. [4]
- (ii)** Use the result in **(i)** to show that if a, b, c and d are positive real numbers, then
- $$(ad^3 + ba^3 + cb^3 + dc^3)(ab + bc + cd + da) \geq abcd(a + b + c + d)^2. \quad [3]$$
- (iii)** Hence show that $(ad^3 + ba^3 + cb^3 + dc^3)(ab + bc + cd + da) \geq 16(abcd)^{\frac{3}{2}}$.
Find a condition relating a, b, c and d for equality to hold, explaining your answer. [4]
- 3 (a)** Find $\int \frac{\cos x}{\sin x - \cos^2 x} dx$. [4]
- (b) (i)** For a twice differentiable function f , if $f(x)$ satisfies the differential equation $f''(x) = \lambda f'(x)f(x)$ for some real constant λ , express $\int [f'(x)]^2 \cdot [f(x)]^N dx$ in terms of $f(x)$, $f'(x)$, λ and N , where N is a positive integer. [4]
- (ii)** Verify that the above result can be applied when $f(x) = \tan 4x$.
Hence find the exact value of $\int_0^{\frac{\pi}{16}} \frac{\sin^{246} 4x}{\cos^{250} 4x} dx$. [4]

- 4 For two integers x and y , let their highest common factor be denoted by $\text{hcf}(x, y)$ and their lowest common multiple by $\text{lcm}(x, y)$.

Given that a and b are positive integers,

- (i) show that there exists integers x and y such that $\text{hcf}(x, y) = a$ and $\text{lcm}(x, y) = b$ if and only if a is a divisor of b . [4]
- (ii) if a is a divisor of b and r is the number of distinct primes in the prime factorisation of $\frac{b}{a}$, show that the number of distinct ordered pairs of positive integers, (x, y) , such that $\text{hcf}(x, y) = a$ and $\text{lcm}(x, y) = b$, is 2^r . [5]

- 5 Happie Newz Café sells 3 types of muffins, namely chocolate, strawberry and vanilla. The café owner bakes a large number of muffins of each type. All muffins of the same flavour are taken to be identical to one another.

- (i) The café owner wants to put 20 muffins in a row on a glass panel so as to entice customers to buy muffins. Find the number of ways she can do so if
- (a) there are no restrictions, [1]
- (b) no 2 muffins of the same type are adjacent. [2]
- (ii) Mr Tay wants some muffins for his 6 students. Find the number of ways that he can do so if
- (a) every student gets at least 1 muffin and no student gets 2 or more muffins of the same type, [2]
- (b) every student gets exactly 4 muffins and no 2 students get the same combination of muffins. [3]
- (iii) At the end of the day, the café owner is left with 6 chocolate, 7 strawberry and 8 vanilla muffins. She wants to throw away these muffins into the 4 different rubbish bins in the vicinity of the café. She wants to ensure that at least 1 muffin is thrown into each bin so as to avoid detection by the members of public that she is wasting food. Find the number of ways she can do so. [5]

- 6 (i) Let $L_{n,k}$ denote the number of ways to distribute n distinct objects into k identical boxes, such that no box is empty and the ordering of objects within each box matters.

(a) Find $L_{n,n}$ and $L_{n,1}$, giving your answer in terms of n whenever possible. [2]

(b) Explain why $L_{n,n-1} = n^2 - n$ for $n \geq 2$. [2]

(c) Show that $L_{n,k} = L_{n-1,k-1} + (n+k-1)L_{n-1,k}$ for $n \geq k \geq 2$. [3]

- (ii) Let $l_{n,k}$ denote the number of ways to distribute n distinct objects into k distinct boxes, such that no box is empty and the ordering of objects within each box matters.

(a) Use part (i)(c) to show that $l_{n,k} = k l_{n-1,k-1} + (n+k-1)l_{n-1,k}$ for $n \geq k \geq 2$. [2]

(b) Explain why $\frac{l_{n,k}}{n!} = \binom{n-1}{k-1}$. [2]

(c) Hence, by using bijective principle, or otherwise, show that for $n \geq k$,

$$\sum_{j=1}^k \binom{k}{j} \binom{n-1}{j-1} = \frac{(n+k-1)!}{n!(k-1)!} \quad [5]$$

- 7 Two sequences $u_1, u_2, u_3, \dots$ and $v_1, v_2, v_3, \dots$ are given by

$$u_1 = 1, \quad v_1 = 1, \quad \text{and}$$

$$u_{n+1} = u_n + 3v_n, \quad v_{n+1} = 2u_n + 7v_n,$$

for positive integers n . The sequence $r_1, r_2, r_3, \dots$ is such that $r_n = \frac{u_n}{v_n}$ for positive integers n .

- (i) Use induction to prove that $2u_n^2 - 3v_n^2 + 6u_nv_n = 5$ for all positive integers n . [4]

It is given that as $n \rightarrow \infty$, $v_n \rightarrow \infty$ and $r_n \rightarrow L$ for some real constant L .

- (ii) Using the results in (i) or otherwise, show that $L = \frac{1}{2}(-3 + \sqrt{15})$. [3]

- (iii) Describe the behaviour of the sequence $r_1, r_2, r_3, \dots$, justifying your answer by considering $r_{n+1} - r_n$. Hence, show that $\sqrt{15}$ lies within the interval $\left(\frac{15}{2r_n + 3}, 2r_n + 3\right)$ for all positive integers n . [4]

- (iv) Deduce that $\frac{213}{55} < \sqrt{15} < \frac{275}{71}$. [2]

- 8** For any positive integer x , $\phi(x)$ is defined to be the number of positive integers not exceeding x which are coprime to x . For example, $\phi(4) = 2$ since positive integers not exceeding 4 and coprime to 4 are 1 and 3.

(i) Find $\phi(p)$ when p is prime. [1]

(ii) Show that, if p is prime and r is a positive integer, then $\phi(p^r) = p^{r-1}(p-1)$. [2]

It is given that $\phi(mn) = \phi(m)\phi(n)$ for positive integers m and n such that $\gcd(m, n) = 1$.

(iii) Show that $\phi(x)$ is even for all $x \geq 3$. [5]

(iv) Find all positive integers x such that $\phi(3x) = \phi(2x)$. [7]

